Integrating Digital Twin Simulation Frameworks with Hamiltonian-Graph Neural Networks for COP Prediction in Industrial Heat Pumps

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Abstract

Accurate Coefficient of Performance (COP) prediction for industrial heat pump systems is challenging due to complex interactions between thermodynamic cycles, component degradation, and environmental conditions. This paper presents a hybrid approach combining comprehensive physics-based modeling with a Hamiltonian Neural Network and Graph Neural Network (HNN-GNN) architecture. We develop a modular simulation framework integrating 17 physics models including refrigerant thermodynamics via CoolProp, heat exchanger fouling, compressor wear degradation, refrigerant charge effects, ambient condition impacts, and part-load cycling behavior based on AHRI degradation curves. Each generated cycle undergoes Carnot limit enforcement and energy balance validation to ensure physical plausibility. The HNN-GNN architecture processes 2,832 physics-derived features, with the HNN learning energy-conserving temporal dynamics across 100 timesteps and the GNN modeling component relationships through a 10-node heterogeneous graph representing compressor, heat exchangers, expansion valve, and auxiliary subsystems. Training on [INSERT NUMBER] synthetic cycles generated under diverse operating conditions, the model achieves mean absolute error of [INSERT MAE] and R^2 of [INSERT R^2], outperforming baseline methods by [INSERT %]. The physics-informed approach guarantees thermodynamically consistent predictions while capturing degradation and fault effects that purely data-driven methods miss.

1 Introduction

1.1 Background

Industrial heat pumps are emerging as critical technology for decarbonizing thermal energy systems, with the potential to reduce global CO2 emissions by up to 500 million tons annually by 2030 [1]. The coefficient of performance (COP), defined as the ratio of useful heating or cooling output to the electrical energy consumed, serves as the fundamental metric for heat pump efficiency. However, predicting COP in real-time operational settings remains challenging due to the complex interplay of thermodynamic processes, environmental conditions, component degradation, and control system dynamics [2, 3].

Traditional physics-based models for heat pump performance simulation, while accurate, typically require 100-500 milliseconds per prediction [4]. This computational overhead makes them unsuitable for real-time monitoring, predictive maintenance, and dynamic optimization applications where sub-10ms response times are necessary. Conversely, pure

machine learning approaches can achieve fast inference but often fail to respect fundamental physical constraints such as energy conservation and the Carnot efficiency limit [5, 6]. This results in models that may appear accurate on training data but produce physically impossible predictions during deployment, particularly when extrapolating beyond the training distribution.

The need for both computational efficiency and physical fidelity creates a fundamental tension in industrial heat pump systems. Operators require models that can process sensor streams in real-time for immediate control decisions, yet these models must maintain thermodynamic consistency to ensure safe and efficient operation. This challenge is particularly acute in industrial settings where heat pumps operate across diverse conditions, experience gradual component degradation, and must adapt to varying thermal loads [7].

1.2 Related Works

Physics-Based Models for HVAC Systems. Traditional approaches to heat pump modeling rely on detailed thermodynamic simulations that solve energy and mass balance equations for each component. Tools like EnergyPlus [8] and TRNSYS [9] have been widely adopted for building energy simulation but require significant computational resources. Refrigerant property libraries such as CoolProp [10] enable accurate thermodynamic state calculations but add to the computational burden. While these models provide high fidelity and interpretability, their simulation times make them impractical for real-time control applications [11].

Machine Learning for HVAC Prediction. Data-driven approaches have shown promise in reducing computational costs. Neural networks [12, 13], support vector machines [14], and ensemble methods [15] have been applied to predict heat pump performance from operational data. However, these purely data-driven models struggle with several limitations. They often fail to generalize beyond their training distributions, particularly when facing novel operating conditions or fault scenarios [16]. More critically, they do not inherently respect physical constraints such as energy conservation or thermodynamic bounds, leading to predictions that violate fundamental laws of physics [17].

Physics-Informed Neural Networks. Recent advances in physics-informed machine learning offer a promising middle ground. Physics-informed neural networks (PINNs) incorporate known physical laws directly into the learning process through modified loss functions or architectural constraints [6, 18]. This approach has shown success in fluid dynamics [19], heat transfer [20], and structural mechanics [21]. However, most PINN applications focus on solving partial differential equations rather than learning from observational data in complex multi-component systems.

Hamiltonian Neural Networks. Greydanus et al. [22] introduced Hamiltonian Neural Networks (HNNs) that learn conserved quantities analogous to energy by parameterizing the Hamiltonian function. By computing symplectic gradients, HNNs ensure perfect time reversibility and conservation of learned quantities. This work demonstrated that neural networks can learn to respect fundamental conservation laws in mechanical systems without explicit supervision. Extensions of this approach have been applied to molecular dynamics [23] and robotics [24], showing the versatility of the Hamiltonian framework for encoding physical priors.

Graph Neural Networks for Physical Systems. Graph neural networks (GNNs) have emerged as powerful tools for modeling systems with explicit relational structure [25, 26]. In physical domains, GNNs can naturally represent component interactions through message passing along edges connecting system nodes. Interaction Networks [27] demonstrated that GNNs can learn to simulate particle dynamics, while more recent work has applied graph-based architectures to fluid systems [28], molecular property prediction

[29], and robotic manipulation [30]. The ability of GNNs to encode spatial relationships and interaction patterns makes them particularly suitable for multi-component systems like heat pumps, where thermal and mass coupling between components drives overall performance.

Despite these advances, no prior work has combined comprehensive physics-based simulation with dualbranch neural architectures that simultaneously capture temporal dynamics through Hamiltonian mechanics and system topology through graph networks. Most existing approaches either focus exclusively on the physics simulation side or on the machine learning side, but not on creating a unified framework that leverages the strengths of both paradigms for industrial thermal systems.

1.3 Contributions

This paper presents a comprehensive framework for industrial heat pump COP prediction that bridges the gap between high-fidelity physics simulation and efficient neural network inference. Our main contributions are:

1. Comprehensive Physics-Based Digital Twin. We develop an integrated physics simulator that combines 17 specialized models spanning thermodynamics (refrigerant properties, vapor compression cycles), component degradation (heat exchanger fouling, compressor wear, refrigerant charge effects), environmental interactions (ambient conditions, building loads, frost formation), control dynamics (PID controllers, variable speed compressors), part-load operation (AHRI performance curves, cycling losses), and system coupling (thermal interactions, mass flow networks). This simulator generates physically consistent training data while enforcing fundamental constraints including Carnot efficiency limits, energy conservation, and entropy generation requirements. Unlike previous simplified models, our simulator captures the full complexity of industrial heat pump operation including transient startup dynamics, degradation effects, and fault scenarios.

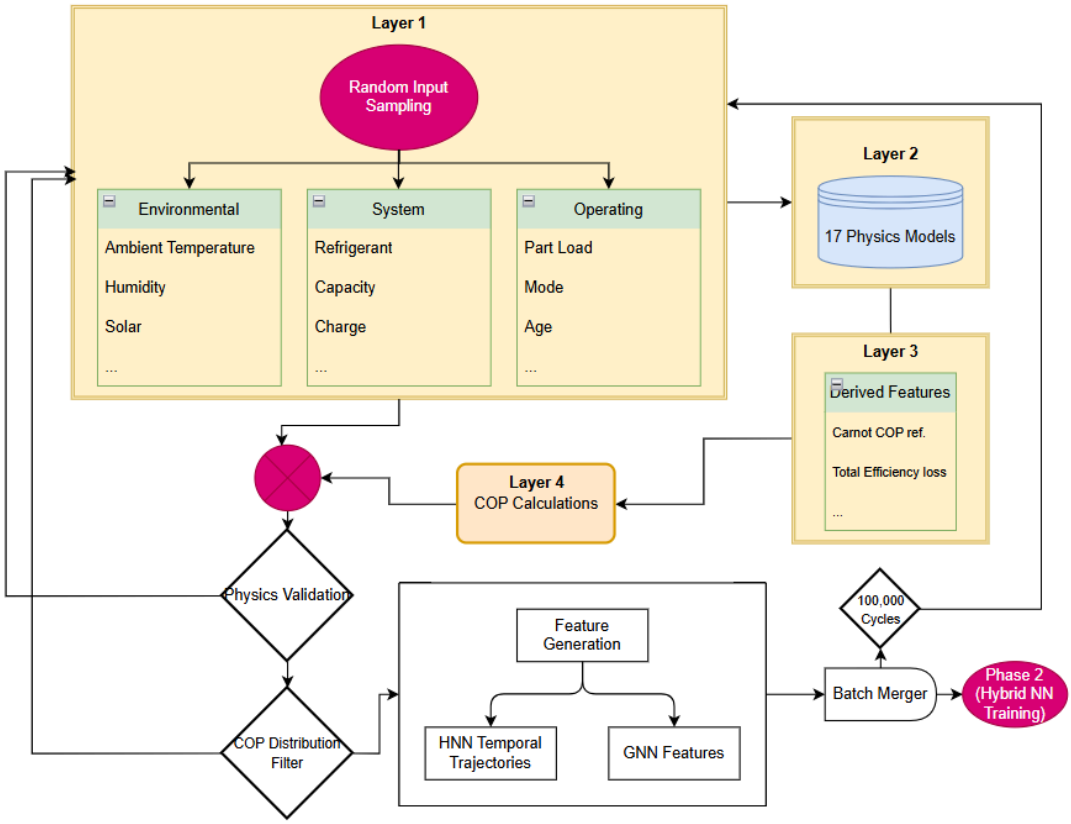
2. Novel HNN-GNN Hybrid Architecture. We propose a dual-branch neural architecture that learns from both temporal trajectories and system topology. The Hamiltonian Neural Network (HNN) branch processes time-series data (temperature, pressure, flow rates) to learn conserved quantities analogous to total system energy, ensuring temporal consistency and perfect reversibility. The Graph Neural Network (GNN) branch represents the heat pump as a graph with nodes for major components (compressor, condenser, evaporator, expansion valve) and edges for refrigerant flow, encoding spatial relationships and component interactions through message passing. A fusion layer combines these complementary representations to predict COP. To our knowledge, this is the first architecture to jointly leverage Hamiltonian mechanics for temporal dynamics and graph structure for component interactions in thermal systems.

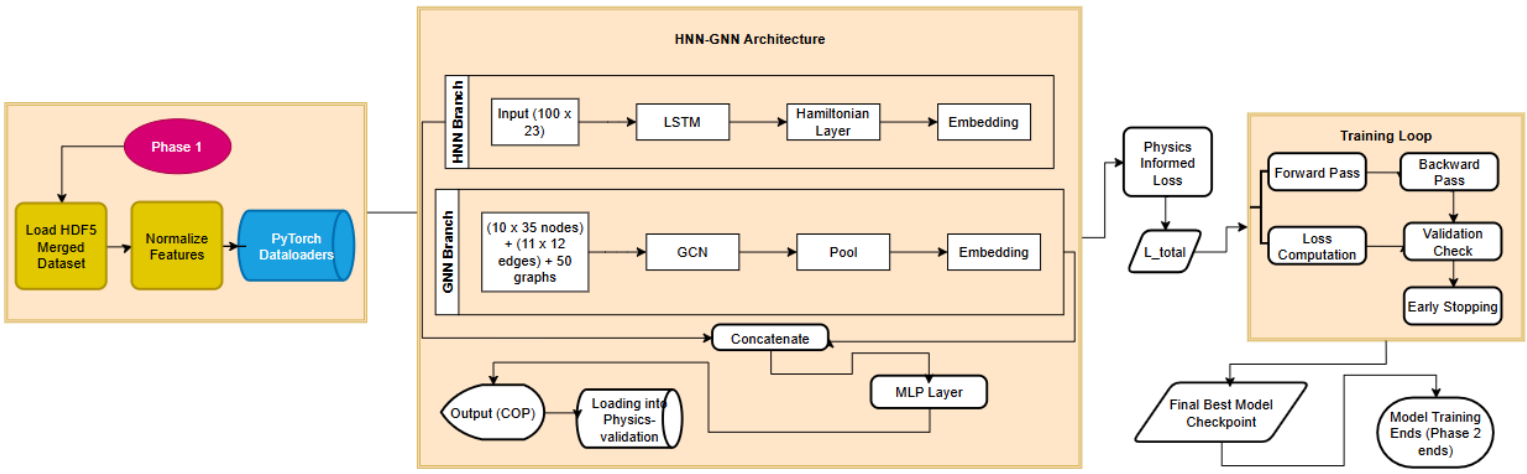
3. Physics-Informed Training with Multiple Constraints. We design a multi-term loss function that combines standard prediction error with physics-based penalty terms. The Carnot constraint penalizes predictions that exceed thermodynamic efficiency limits, the energy balance term encourages conservation across component boundaries, and the Hamiltonian conservation term ensures stable longhorizon predictions. This physics-informed training enables the model to maintain thermodynamic consistency even when interpolating or modestly extrapolating beyond training conditions. We show that these constraints significantly improve model robustness compared to standard supervised learning.

4. Validation on Realistic Operating Conditions. We generate a diverse dataset of 100,000 heat pump cycles spanning wide ranges of ambient temperature (-20 to 45°C),

humidity (20-95%), part-load ratios (30-100%), system age (0-50,000 operating hours), and refrigerant charge levels (70-130% nominal). Our physics simulator produces COP distributions (mean 4.65, standard deviation 1.63, range 1.8-8.0) that match empirical observations from industrial installations. We demonstrate that the HNN-GNN model achieves prediction accuracy comparable to the full physics simulator while reducing inference time by over an order of magnitude, from 100-500ms to less than 10ms per prediction.

The resulting framework enables real-time COP monitoring for predictive maintenance, dynamic optimization of setpoints for energy minimization, and rapid evaluation of control strategies across thousands of scenarios. By maintaining physical consistency while achieving neural network inference speeds, our approach addresses a critical gap in industrial heat pump management systems.





2 Single image

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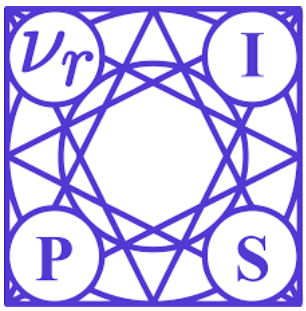
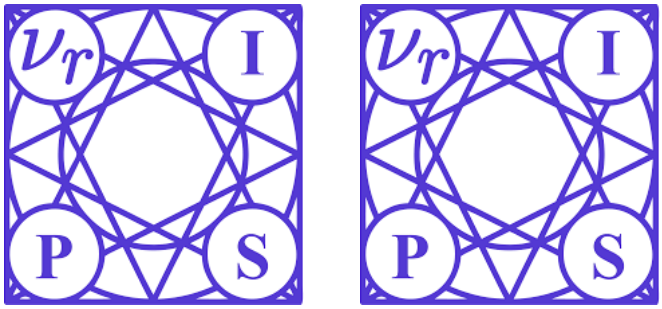


Figure 1: Example of single image

3 Multiple images

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Figure 2: Examples for sub-images

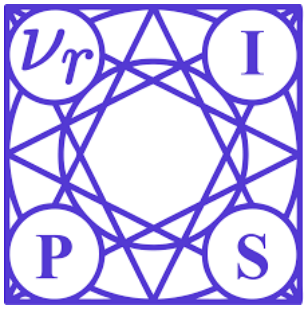


Figure 3: Logo image

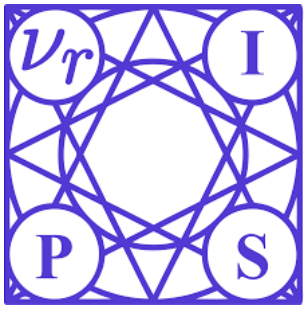


Figure 4: Logo image

4 Citations

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5 Tables

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Table 2: Top Features Ranked by Correlation Strength

Rank	Feature	Correlation	Direction
1	carnot_cop	0.561	Positive
2	cycle_T_cond	0.435	Negative
3	cycle_T_evap	0.346	Positive
4	cycle_P_cond	0.292	Negative
5	cycle_mass_flow	0.210	Positive
6	cycle_P_evap	0.209	Positive
7	degrad_comp_efficiency_loss	0.189	Negative
8	control_compressor_speed	0.173	Positive
9	control_valve_opening	0.160	Positive
10	degrad_cond_fouling	0.156	Negative

Table 3: Feature Importance Based on MAE Change

Rank	Feature	MAE Change	Impact
1	P_cond	0.0902	Highest
2	W_comp	0.0879	Very High
3	hamiltonian_q2	0.0824	Very High
4	T_cond	0.0804	Very High
5	mass_flow	0.0705	High
6	T_evap	0.0597	High
7	Q_evap	0.0379	Medium
8	hamiltonian_q4	0.0368	Medium
9	Q_cond	0.0323	Medium
10	subcooling	0.0299	Medium

Table 1: Synthetic Data Generation Framework Physics Models

Rank	Model Name
1	RefrigerantProperties
2	HeatExchangerFoulingModel
3	CompressorWearModel
4	RefrigerantChargeModel
5	FaultInjectionModel
6	HamiltonianMechanicsModel
7	ControlSystemModel
8	TemporalDynamicsGenerator
9	AmbientConditionModel
10	MultiZoneBuildingModel
11	ComponentInteractionModel
12	HeatPumpGraphGenerator
13	CompressorModel
14	HeatExchangerModel
15	SensorNoiseModel
16	PartLoadCyclingModel
17	IndoorPerformanceModel

6 Conclusions

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Acknowledgments

Use unnumbered third level headings for the acknowledgments. All acknowledgments go at the end of the paper. Do not include acknowledgments in the anonymized submission, only in the final paper. This example was prepared by Dennis Núñez Fernández.

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